

1 USV and controller parameters

First, given the matrices A_η , A_ν and B_{uw} as:

$$\begin{aligned} A_\eta &= -M^{-1}G = \begin{bmatrix} -0.0358 & 0 & 0 \\ 0 & -0.0208 & 0 \\ 0 & -0.0394 & 0 \end{bmatrix}, \\ A_\nu &= -M^{-1}N = \begin{bmatrix} -0.0797 & 0 & 0 \\ 0 & -0.0818 & -0.0577 \\ 0 & -0.2254 & -0.2535 \end{bmatrix}, \\ B_{uw} &= M^{-1} = \begin{bmatrix} 0.9215 & 0 & 0 \\ 0 & 0.7802 & 1.4811 \\ 0 & 1.4811 & 7.4562 \end{bmatrix}. \end{aligned}$$

Accordingly, for $\theta(t) \in [-\frac{\pi}{6}, \frac{\pi}{6}]$, the nonlinear USV can be modeled by the subsequent local T-S fuzzy systems:

$$\begin{aligned} A_i &= \begin{bmatrix} 0 & J_i \\ A_\eta & A_\nu \end{bmatrix}, \quad B_i = \begin{bmatrix} 0 \\ B_{uw} \end{bmatrix}, \\ C_{2i} &= [0 \quad 0 \quad -0.002 \quad 0 \quad 0 \quad 0.01], i \in \{1, 2, 3, 4\} \end{aligned}$$

where the corresponding rotational matrices are defined as:

$$\begin{aligned} J_1 &= \begin{bmatrix} 1 & -1/2 & 0 \\ 1/2 & -1/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, J_2 = \begin{bmatrix} \sqrt{3}/2 & -1/2 & 0 \\ 1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \\ J_3 &= \begin{bmatrix} 1 & 1/2 & 0 \\ -1/2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, J_4 = \begin{bmatrix} \sqrt{3}/2 & 1/2 & 0 \\ -1/2 & \sqrt{3}/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \end{aligned}$$

Furthermore, the FMF are explicitly constructed as:

$$\begin{aligned} g_1(\theta(t)) &= \phi_1(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_2(\theta(t)) &= \phi_1(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_3(\theta(t)) &= \phi_2(x_3(t))(-3 - 2\sqrt{3} + (4 + 2\sqrt{3})\cos(x_3(t))), \\ g_4(\theta(t)) &= \phi_2(x_3(t))(4 + 2\sqrt{3} - (4 + 2\sqrt{3})\cos(x_3(t))), \end{aligned}$$

in which the intermediate variables ϕ_1 and ϕ_2 are given by:

$$\phi_1(x_3(t)) = \frac{1}{2} + \sin(x_3(t)), \quad \phi_2(x_3(t)) = \frac{1}{2} - \sin(x_3(t)).$$

First, the system matrices A_η , A_ν , and B_{uw} in (4) are adopted from Ye et al. Following the wave superposition approach in Hong et al, the disturbance $w(t) = [w_1(t), w_2(t), w_3(t)]^T$ is constructed using the Pierson–Moskowitz (PM) spectrum:

$$w_d(t) = \sum_{q=1}^{50} [a_d H_q \cos(\omega_q t + \epsilon_q + p_d) + b_d H_q^2], d = 1, 2, 3,$$

where $H_q = \sqrt{2S(\omega_q)\Delta\omega}$, $S(\omega_q)$ is the PM spectrum, and ϵ_q is uniformly distributed over $[0, 2\pi]$. For small USVs operating in mild nearshore seas, $H_s = 0.30$ m is selected, with phase offsets $p_1 = 0$, $p_2 = 0.5$, and $p_3 = 1$. The channel coefficients are selected as

$$a_d = \frac{4\sigma_d}{H_s}, b_d = \frac{8\beta\sigma_d}{H_s^2},$$

where $\sigma_1 = 0.010$, $\sigma_2 = 0.008$, and $\sigma_3 = 0.003$ are the nominal RMS amplitudes, and $\beta = 0.05$ sets the relative mean drift. The encounter and wave frequencies are taken equal.

The feedback control gains alongside the triggering weight matrices are computed as follows:

$$K_{11} = \begin{bmatrix} -0.0916 & -0.0050 & -0.0006 & -0.6106 & 0.0008 & -0.0017 \\ 0.0084 & -0.1693 & 0.0202 & -0.0443 & -1.1410 & 0.2297 \\ 0.0007 & 0.0289 & -0.0241 & 0.0153 & 0.2205 & -0.1219 \end{bmatrix},$$

$$K_{12} = \begin{bmatrix} -0.0728 & -0.0040 & -0.0001 & -0.4846 & -0.0046 & 0.0016 \\ 0.0035 & -0.1324 & 0.0152 & -0.0478 & -0.8923 & 0.1778 \\ 0.0013 & 0.0232 & -0.0174 & 0.0166 & 0.1755 & -0.0918 \end{bmatrix},$$

$$K_{13} = \begin{bmatrix} -0.0498 & -0.0028 & 0.0000 & -0.3313 & -0.0055 & 0.0027 \\ -0.0001 & -0.0903 & 0.0098 & -0.0461 & -0.6079 & 0.1182 \\ 0.0015 & 0.0160 & -0.0112 & 0.0148 & 0.1203 & -0.0606 \end{bmatrix},$$

$$K_{21} = \begin{bmatrix} -0.0963 & -0.0075 & -0.0014 & -0.6414 & 0.0013 & -0.0028 \\ 0.0157 & -0.1902 & 0.0215 & -0.0150 & -1.2467 & 0.2479 \\ -0.0029 & 0.0369 & -0.0249 & 0.0020 & 0.2525 & -0.1363 \end{bmatrix},$$

$$K_{22} = \begin{bmatrix} -0.0783 & -0.0051 & -0.0017 & -0.5216 & 0.0015 & -0.0022 \\ 0.0121 & -0.1515 & 0.0173 & -0.0029 & -0.9926 & 0.2001 \\ -0.0029 & 0.0290 & -0.0199 & -0.0052 & 0.1994 & -0.1113 \end{bmatrix},$$

$$K_{23} = \begin{bmatrix} -0.0551 & -0.0031 & -0.0015 & -0.3675 & 0.0014 & -0.0015 \\ 0.0088 & -0.1039 & 0.0122 & 0.0064 & -0.6806 & 0.1397 \\ -0.0028 & 0.0195 & -0.0140 & -0.0099 & 0.1354 & -0.0790 \end{bmatrix},$$

$$K_{31} = \begin{bmatrix} -0.0916 & 0.0049 & 0.0005 & -0.6102 & -0.0011 & 0.0018 \\ -0.0084 & -0.1694 & 0.0203 & 0.0448 & -1.1423 & 0.2300 \\ -0.0006 & 0.0290 & -0.0241 & -0.0154 & 0.2210 & -0.1220 \end{bmatrix},$$

$$K_{32} = \begin{bmatrix} -0.0727 & 0.0040 & 0.0001 & -0.4841 & 0.0047 & -0.0016 \\ -0.0035 & -0.1324 & 0.0153 & 0.0485 & -0.8933 & 0.1781 \\ -0.0013 & 0.0233 & -0.0174 & -0.0168 & 0.1759 & -0.0919 \end{bmatrix},$$

$$K_{33} = \begin{bmatrix} -0.0497 & 0.0028 & -0.0000 & -0.3309 & 0.0056 & -0.0026 \\ 0.0001 & -0.0904 & 0.0099 & 0.0469 & -0.6091 & 0.1187 \\ -0.0015 & 0.0161 & -0.0112 & -0.0150 & 0.1208 & -0.0607 \end{bmatrix},$$

$$K_{41} = \begin{bmatrix} -0.0963 & 0.0075 & 0.0014 & -0.6414 & -0.0013 & 0.0028 \\ -0.0157 & -0.1902 & 0.0215 & 0.0150 & -1.2466 & 0.2480 \\ 0.0029 & 0.0369 & -0.0249 & -0.0021 & 0.2525 & -0.1363 \end{bmatrix},$$

$$K_{42} = \begin{bmatrix} -0.0783 & 0.0051 & 0.0017 & -0.5216 & -0.0015 & 0.0022 \\ -0.0121 & -0.1515 & 0.0173 & 0.0030 & -0.9925 & 0.2001 \\ 0.0029 & 0.0290 & -0.0199 & 0.0052 & 0.1994 & -0.1113 \end{bmatrix},$$

$$K_{43} = \begin{bmatrix} -0.0551 & 0.0031 & 0.0015 & -0.3675 & -0.0014 & 0.0015 \\ -0.0088 & -0.1039 & 0.0122 & -0.0064 & -0.6805 & 0.1397 \\ 0.0027 & 0.0195 & -0.0140 & 0.0099 & 0.1354 & -0.0790 \end{bmatrix}.$$

$$\Psi_1 = \begin{bmatrix} 0.0243 & 0.0002 & -0.0004 & 0.1572 & -0.0013 & -0.0025 \\ 0.0002 & 0.0185 & -0.0002 & 0.0087 & 0.1206 & -0.0165 \\ -0.0004 & -0.0002 & 0.0029 & -0.0014 & -0.0051 & 0.0122 \\ 0.1572 & 0.0087 & -0.0014 & 1.0441 & 0.0383 & -0.0195 \\ -0.0013 & 0.1206 & -0.0051 & 0.0383 & 0.8060 & -0.1262 \\ -0.0025 & -0.0165 & 0.0122 & -0.0195 & -0.1262 & 0.0698 \end{bmatrix},$$

$$\Psi_2 = \begin{bmatrix} 0.0255 & -0.0004 & 0.0007 & 0.1679 & -0.0086 & 0.0003 \\ -0.0004 & 0.0208 & -0.0005 & 0.0051 & 0.1326 & -0.0192 \\ 0.0007 & -0.0005 & 0.0034 & 0.0052 & -0.0060 & 0.0156 \\ 0.1679 & 0.0051 & 0.0052 & 1.1185 & -0.0098 & -0.0028 \\ -0.0086 & 0.1326 & -0.0060 & -0.0098 & 0.8587 & -0.1356 \\ 0.0003 & -0.0192 & 0.0156 & -0.0028 & -0.1356 & 0.0904 \end{bmatrix},$$

$$\Psi_3 = \begin{bmatrix} 0.0242 & -0.0002 & 0.0004 & 0.1571 & 0.0013 & 0.0025 \\ -0.0002 & 0.0185 & -0.0002 & -0.0088 & 0.1203 & -0.0165 \\ 0.0004 & -0.0002 & 0.0029 & 0.0014 & -0.0051 & 0.0121 \\ 0.1571 & -0.0088 & 0.0014 & 1.0426 & -0.0389 & 0.0193 \\ 0.0013 & 0.1203 & -0.0051 & -0.0389 & 0.8050 & -0.1261 \\ 0.0025 & -0.0165 & 0.0121 & 0.0193 & -0.1261 & 0.0696 \end{bmatrix},$$

$$\Psi_4 = \begin{bmatrix} 0.0255 & 0.0004 & -0.0007 & 0.1678 & 0.0086 & -0.0003 \\ 0.0004 & 0.0208 & -0.0005 & -0.0051 & 0.1326 & -0.0192 \\ -0.0007 & -0.0005 & 0.0034 & -0.0052 & -0.0060 & 0.0156 \\ 0.1678 & -0.0051 & -0.0052 & 1.1185 & 0.0098 & 0.0028 \\ 0.0086 & 0.1326 & -0.0060 & 0.0098 & 0.8586 & -0.1356 \\ -0.0003 & -0.0192 & 0.0156 & 0.0028 & -0.1356 & 0.0904 \end{bmatrix}.$$